Fiscal Year:	FY 2020	**Task Last Updated:**	FY 09/17/2020
PI Name:	DeLucas, Lawrence J. O.D., Ph.D.		
Project Title:	The Effect of Macromolecular Transport on Microgravity Protein Crystallization		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	BIOPHYSICS--Biophysics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90245	**Congressional District:**	33
Comments:	NOTE: PI retired in 2017 from the University of Alabama, Birmingham, and is now Principal Scientist at The Aerospace Corporation, El Segundo, CA (Ed., 5/22/2018)		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	10/18/2017	**End Date:**	07/31/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Penn, Benjamin	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 7/31/2020 per NSSC information (Ed., 8/12/20) NOTE: New follow-on grant awarded, 80NSSC19K1268, per NSSC information. New end date is 6/30/2020 (Ed., 1/30/2020) NOTE: Extended to 2/22/2019 per NSSC information (Ed., 10/16/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Betzel, Christian D.Sc. (University of Hamburg, Germany)		
Grant/Contract No.:	80NSSC19K1268 ; 80NSSC18K0013		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "The Effect of Macromolecular Transport on Microgravity Protein Crystallization," grant NNX13AQ23G, when Principal Investigator Lawrence DeLucas was at University of Alabama, Birmingham. PI is now Principal Scientist with The Aerospace Corporation. Proteins are important macromolecules without which our bodies would be unable to repair, regulate, or protect themselves. Determination of the atomic three-dimensional structure of proteins provides critical information that allows scientists to understand how they function. It also provides critical structural information to support the development of new pharmaceuticals to treat chronic and infectious diseases. X-ray crystallography is used to determine protein structure but first requires the growth of well-ordered protein crystals that are of sufficient quality. Protein crystals previously grown in microgravity have resulted in more than fifty examples of clear improvements in crystal quality via x-ray diffraction analysis. The majority of the crystal improvements resulted in improved resolution of the three-dimensional protein structures as determined via x-ray crystallography. Two theories have been proposed to explain the observed microgravity crystal improvements. Both are based on protein transport rate differences in microgravity versus unit gravity environments. One theory is based on expected slower crystal growth rates, due to slower protein transport to the growing crystal surface. The second theory proposes that there is an exclusion of larger protein aggregates becoming incorporated into the growing crystal (aggregates produce crystal defects that reduce crystal quality). The theoretical explanation is based on the expected faster transport of single protein molecules as opposed to aggregates of two or more protein molecules. Unfortunately, there is insufficient experimental data to validate either theory. This proposal will utilize different Light Microscopy Module (LMM) optical systems to determine which theory, via a series of experiments, addresses the void in our understanding of microgravity's effect on growing protein crystals.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ultimate goal of this research is to determine the mechanism by which microgravity affects the quality of protein crystals. The research will utilize The Light Microscopy Module (LMM) to investigate two possible theories for the improved crystal quality observed in a microgravity environment: 1) Slower crystal growth rates result in protein that is more perfectly ordered in the crystalline lattice and/or 2) a microgravity environment basically filters out larger aggregates of protein thereby allowing a more uniform population of single protein molecules to become incorporated into the crystalline lattice, thereby producing higher quality crystals. Understanding the mechanism by which microgravity produces higher quality crystals can lead to new ideas for Earth-based crystallization systems that may also improve the quality of protein crystals. This will benefit the structural biology community that determines protein structures to understand how these complicated molecules function in our bodies, in bacteria, and viruses. In many cases protein crystals are not of sufficient quality to enable determination of their structures using a technique called x-ray crystallography. Therefore, new applications that may improve the quality of protein crystals will be of significant benefit as scientists continue to try to understand the function of these complicated molecules.
Task Progress:	Proteins are important macromolecules without which our bodies would be unable to repair, regulate, or protect themselves. The use of X-ray crystallography to determine protein structure requires the production of well-ordered protein crystals that are of sufficient quality. Without high-quality crystals of a protein, it is impossible to carry out crystallographic structural studies. Using three-dimensional structure information, researchers can determine how proteins function and in cases where these proteins are involved in disease processes, the structure is often used to design new drugs that specifically interact with the protein. Although there have been many examples where the microgravity environment yielded protein crystals of superior quality compared to their laboratory (1G) counterparts, there is a lack of understanding as to the reason for improved crystal quality in microgravity. The purpose of this proposal is to address the two existing hypotheses for the observed improved macromolecular crystallization in a microgravity environment. This proposal addresses the following hypothesis: Improved quality of microgravity-grown protein crystals is the result of two macromolecular characteristics that exist in a buoyancy-free, diffusion-dominated solution: • Slower crystal growth rates, due to slower protein transport to the growing crystal surface • Predilection of growing crystals to incorporate protein monomers versus higher protein aggregates due to differences in transport rates Significance of Proposed Studies: This proposal will address the void in our fundamental understanding of microgravity protein transport and how it influences both crystal growth rate and crystal quality as a result of differences in macromolecular transport rates for 1g versus µg and differences in the homogeneity of protein (monomers versus larger aggregates) incorporated into the crystalline lattice. In microgravity, an estimate of transport rates for macromolecules is as follows. The hypothesis is addressed in several ways as outlined in the three specific aims shown below: Specific Aims: 1. On the International Space Station (ISS) and concurrently in Earth control experiments, compare incorporation of protein aggregates (monomers versus dimers, trimers, etc.) into growing protein crystals. Detection of aggregates in the crystal will be qualitatively assessed via LMM's and laboratory fluorescence confocal microscopes. Based on the percentage incorporation of larger aggregates within the crystal, the effect of molecular filtering based on differences in diffusion rates will be assessed. 2. Measure crystal growth rates in 1g versus µg for different size aggregates of proteins with molecular weights ranging from less than 20 kDa to more than 1,200 kDa. 3. Compare the defect density and crystal quality via mass spectroscopy and x-ray diffraction quality of crystals grown at different rates in a 1g environment. More than twenty fluorescent data sets and eleven X-ray data sets have been collected for the microgravity-grown lysozyme crystals. Data for an equal number of 1G-grown control crystals are currently being collected. Similar data collection is underway for the glutathione-s-transferase crystals. Conclusions: Comparison of growth rate and fluorescence intensity for the microgravity-grown crystals versus their laboratory control counterparts suggested that: • Protein crystals nucleate and grow more slowly in microgravity.

	• The fluorescence intensity of microgravity grown crystals is markedly lower than their laboratory control counterparts. • Under microgravity conditions larger crystal dimensions for glutathione-S-transferase dimer/tetramer ratios were observed. • Initial results reveal nucleation/growth kinetics and mass transport around growing crystals have a significant effect on the crystal growth rate and final crystal size. • Crystals grown under microgravity for lysozyme and glutathione-S-transferase revealed lower fluorescence intensities then 1G control crystals. • The data obtained suggests that macromolecular mass transport, nucleation kinetics, and crystal growth rates are different in microgravity compared to 1G control experiments and that one or more of these differences may contribute to differences observed in crystal quality using x-ray diffraction analysis.
Bibliography Type:	Description: (Last Updated: 06/29/2023)
Articles in Peer-reviewed Journals	Martirosyan A, DeLucas LJ, Schmidt C, Perbandt M, McCombs D, Cox M, Radka C, Betzel C. "Effect of macromolecular mass transport in microgravity protein crystallization." Gravitational and Space Research. 2019;7:33-44. Published online: 10 Sep 2019. https://doi.org/10.2478/gsr-2019-0005 , Sep-2019
Articles in Peer-reviewed Journals	Martirosyan A, Falke S, McCombs D, Cox M, Radka CD, Knop J, Betzel C, DeLucas LJ. "Tracing transport of protein aggregates in microgravity versus unit gravity crystallization." npj Microgravity. 2022 Feb 17;8:4. https://doi.org/10.1038/s41526-022-00191-x ; PMID: 35177635; PMCID: PMC8854672 , Feb-2022